Rahul Hirondi

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EXPERIENCE

Jet2 Travel Technologies | Data Scientist
July 2024 – Present

- **Real-Time Data Engineering**: Architected a high-throughput pipeline in **Snowflake** using Tasks/Streams to parse nested JSON from AWS WhatsApp transcripts. Enabled millisecond-latency event detection by transitioning from hourly batch processing to real-time streaming.
- **Advanced Prompt Engineering**: Designed and version-controlled complex system prompts utilizing **Context Injection**, **Few-Shot Learning**, and **Chain-of-Thought** reasoning within Snowflake Cortex. Successfully reduced hallucination rates when classifying unstructured chat data.
- **Hybrid Anomaly Detection**: Developed a robust detection engine combining statistical logic (Z-Scores, 98th Percentile) with NLP (Lemmatization, TF-IDF). This dictionary-backed rule engine filtered noise and identified true operational disruptions with **98% precision**.
- **Full-Stack GenAI Application**: Built a production-grade **Streamlit** app with **RBAC**, custom HTML/CSS UI, and Human-in-the-Loop (HITL) feedback mechanisms for model validation and retraining.
- **Operational Impact & Integration**: Configured **Snowflake-to-Teams** integration for real-time alerts. Empowered Ops teams to handle disruptions proactively, reducing support calls by **30%**. Managed MLOps lifecycle and optimized token costs.

Cognizant | Data Scientist
Apr. 2021 – July 2024

- **GenAI Tax Assistant (LLM/RAG)**: Architected a RAG-based conversational AI using **Llama-3** and **LangChain**. Engineered effective instruction-based prompts using **Context Injection** and **Zero-Shot** techniques to guide responses. Integrated **Groq** for high-speed inference and **FAISS** for efficient document retrieval, serving the model via a **Streamlit** UI.
- **MLOps & Pipeline Architecture**: Spearheaded the development of scalable **SQL/Python** pipelines to process large datasets, resulting in a **70% improvement** in processing speed. Automated the lifecycle monitoring of **25+ ML models**, generating actionable insights that reduced manual reporting overhead by **50%**.
- **Fraud Detection System**: Developed a high-precision fraud detection system using ensemble methods (XGBoost, Random Forest). Implemented advanced data standardization and class balancing techniques (**SMOTE**, **ADASYN**) to handle imbalanced transaction data, achieving **92% accuracy** through rigorous hyperparameter tuning.

SKILLS

Languages & Web : Python, SQL, HTML, CSS, JSON
GenAI & LLMs : RAG, Llama-3, Snowflake Cortex, HuggingFace, Prompt Engineering (Few-Shot/CoT), LangChain
Cloud & Tools : Snowflake (Tasks/Streams), AWS, Power BI, Git, APIs (REST), Docker, Streamlit
Machine Learning : NLP (Spacy, NLTK), Deep Learning (TensorFlow, PyTorch), Predictive Modeling, Anomaly Detection
Libraries : Pandas, NumPy, Scikit-Learn, FAISS, Matplotlib, Seaborn, LangGraph

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals (AI-900)
Specialization: NLP using Transformers

EDUCATION

P.G. Diploma in Data Science | IIIT Bangalore — Aug 2020

B.E. Computer Science | CSIT Durg — Aug 2019

ACHIEVEMENTS

Honors & Awards

- **Cognizant GenC Digital Honors**: Recognized for excellence in delivering high-impact AI/ML solutions.
- **V-Force Award**: Awarded for outstanding contribution and innovation in project deliverables at Cognizant.